Editor concern: recent ML baselines

Editor

“to compare the results of the proposed approach with important baselines, including a physics-based one and recent ML ones.”

Recent ML baselines

T/S/u/v: depth-averaged over 2.5–550 m

RMSE diagnostic	Samudra 2	Neptune
Temperature (°C)	0.4287	0.6928
Salinity (psu)	0.1398	0.1764
Zonal u (m/s)	0.03831	0.05366
Meridional v (m/s)	0.03737	0.03804
SSH (cm)	4.95	7.75

Against OISST (2015–2022)

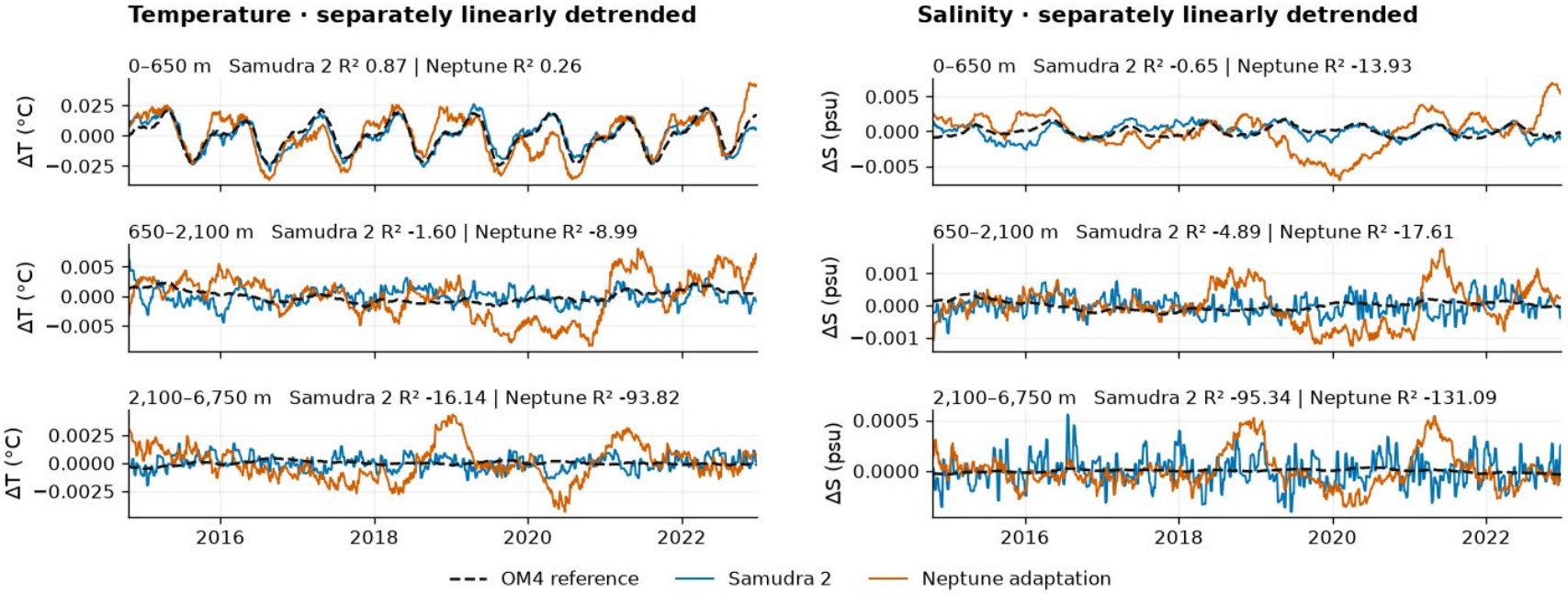
RMSE diagnostic	OM4	Samudra 2
SST (°C)	0.805	0.936

Samudra 2 has lower field errors than the Neptune.

Donno, Davide, et al. "Neptune: An AI model for Global Ocean Subseasonal Prediction." *arXiv preprint arXiv:2609.08606* (2026).

ML baselines: detrended global means

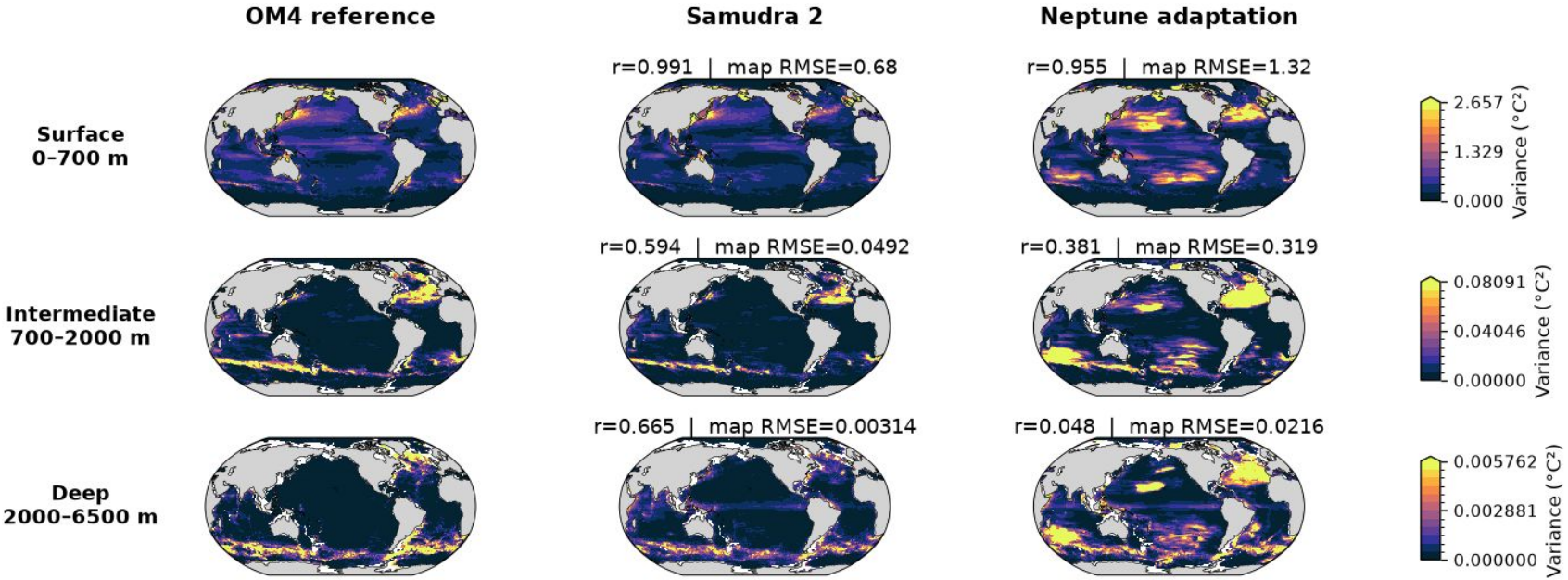
Samudra 2 better tracks detrended global means than Neptune, but deep-ocean errors remain large.



The temporal advantage extends beyond average error; deep-ocean fidelity remains limited.

ML baselines: temperature variance by depth

Samudra 2 better matches patterns.



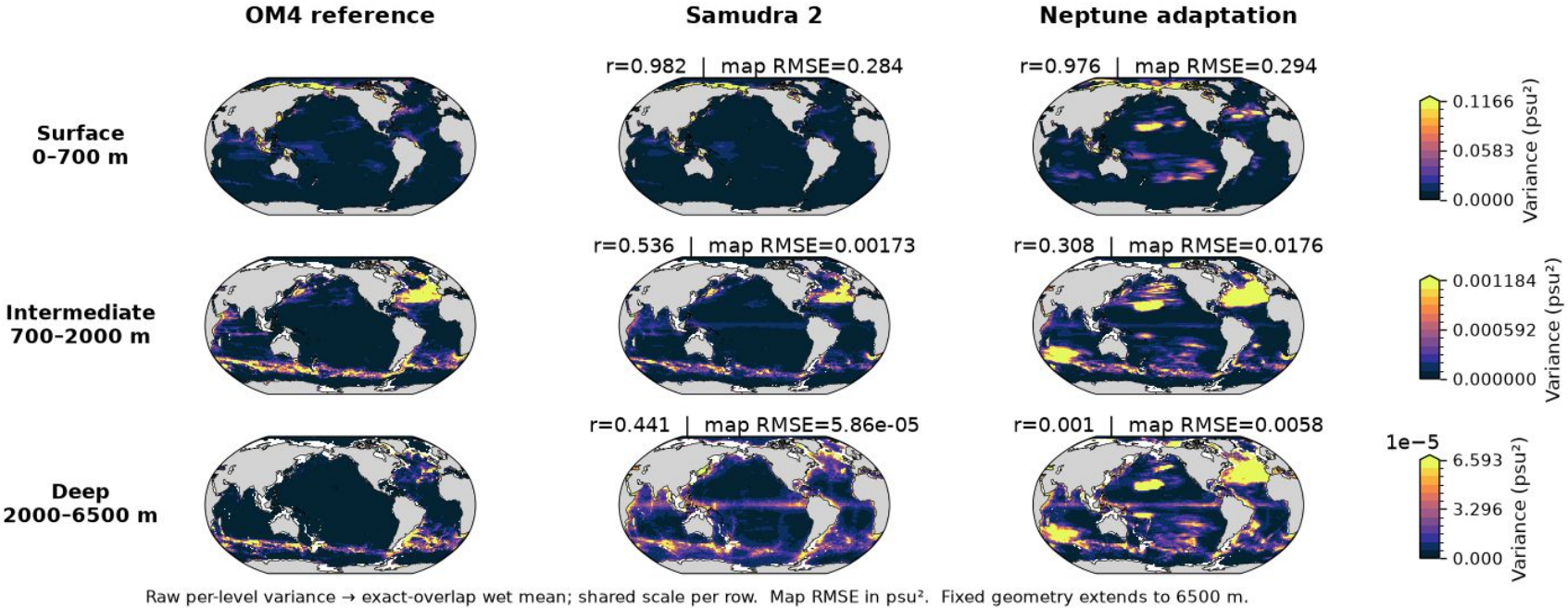
Raw per-level variance → exact-overlap wet mean; shared scale per row. Map RMSE in °C². Fixed geometry extends to 6500 m.

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Samudra 2's advantage extends to spatial variability.

ML baselines: salinity variance by depth

Samudra 2 better matches salinity-variance patterns, but deep excess remains.



Deep-salinity variability remains a weakness despite the improved baseline comparison.

Reviewer concern: changed-forcing generalization

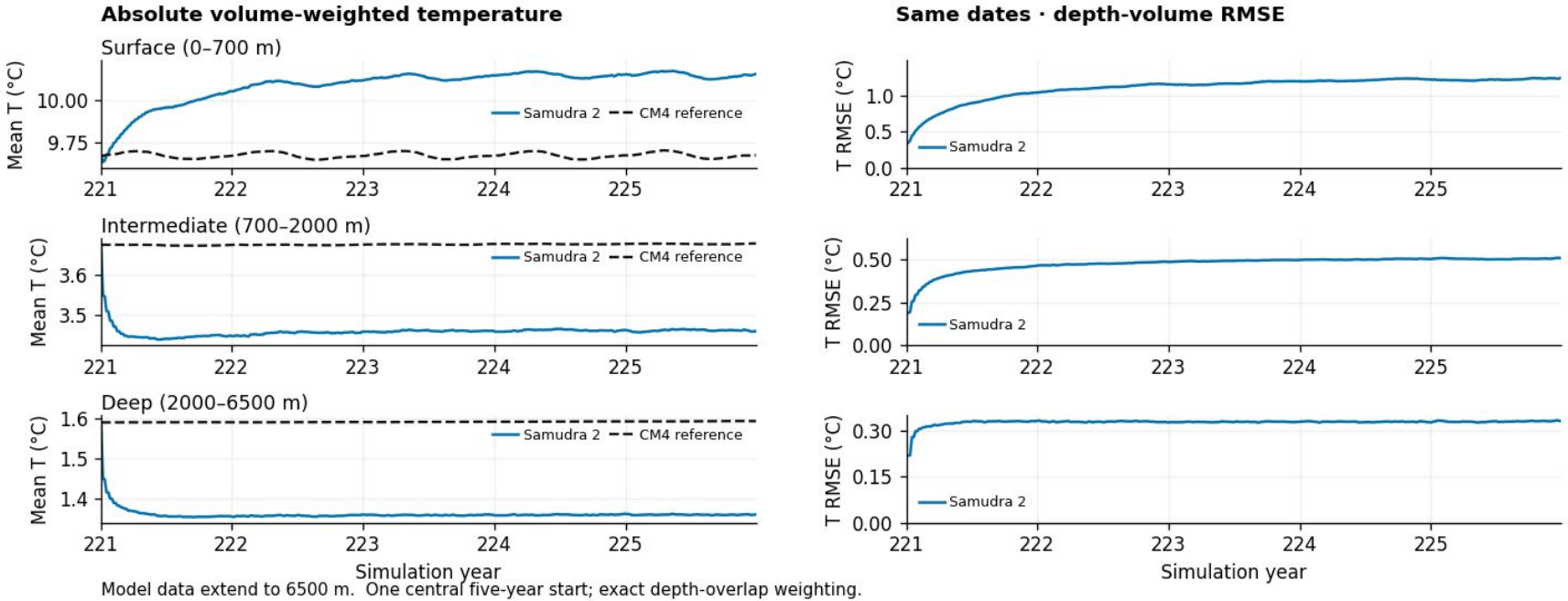
Reviewer 3 · Major comment 4

“It’s unclear whether this approach would work if any forcing term was changed (e.g., using atmospheric forcing from a different CO2 scenario).”

“It is understandable that the community focus on in-distribution performance before moving to stronger generalisation cases, but this limitation should be better highlighted in the paper.”

CM4 piControl: temperature by depth

Samudra 2 retains seasonal upper-ocean variation but develops subsurface cold biases.

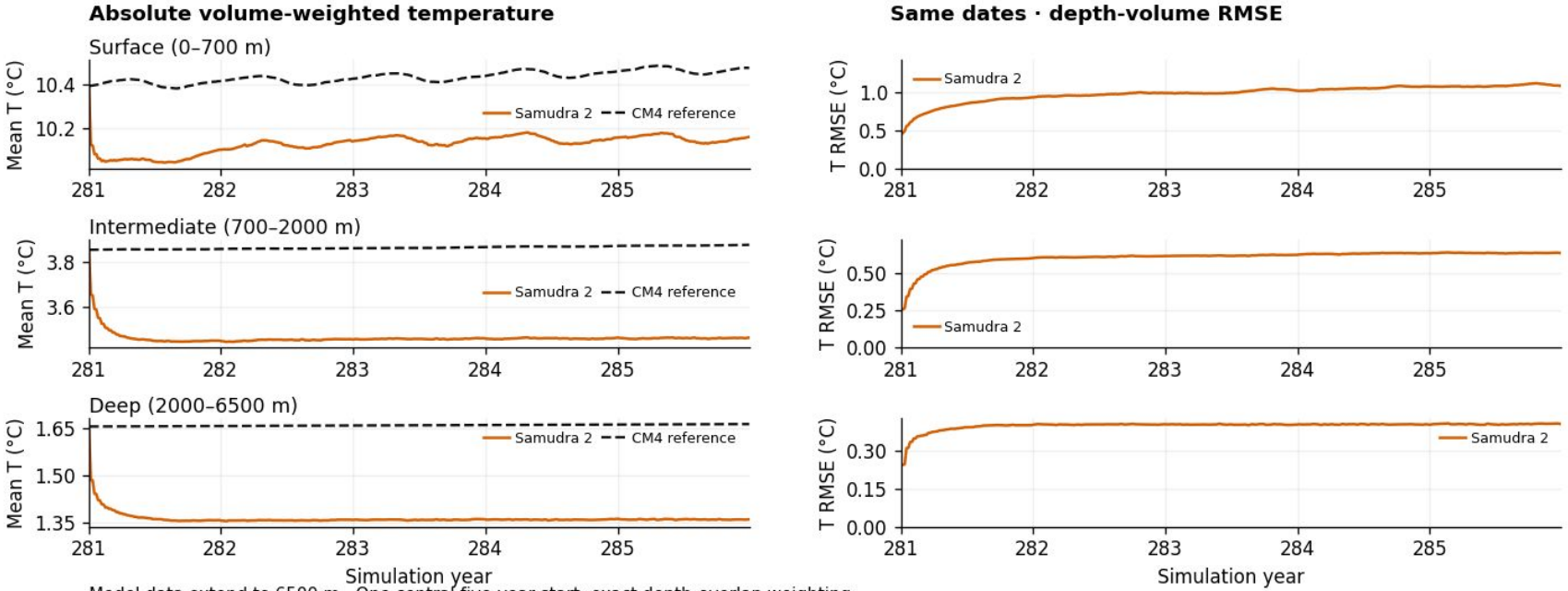


Model data extend to 6500 m. One central five-year start; exact depth-overlap weighting.

The model runs without retraining under CM4 forcing, with persistent subsurface temperature biases.

CM4 1pctCO₂: temperature by depth

In the 1pctCO₂ case, Samudra 2 remains too cold below 700 m.

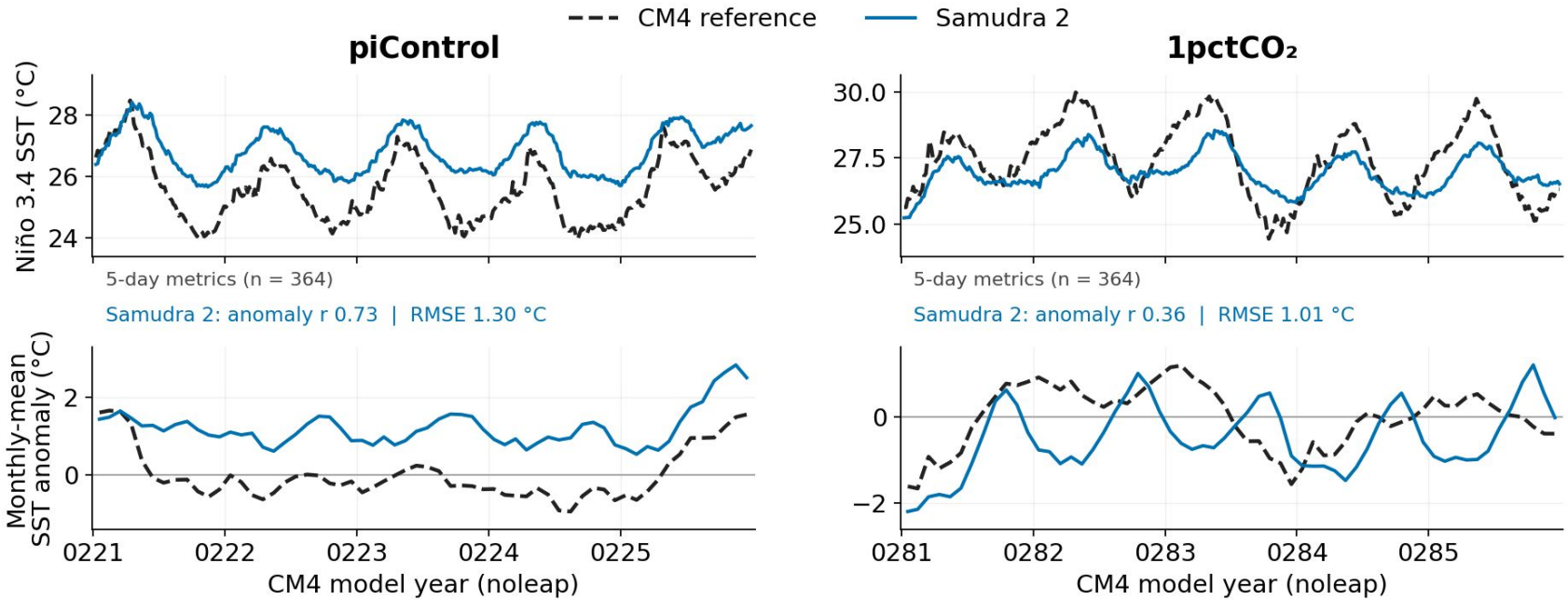


Model data extend to 6500 m. One central five-year start; exact depth-overlap weighting.

The subsurface cold bias limits zero-shot transfer to the increasing-CO₂ scenario.

CM4 zero-shot: Niño 3.4 variability

Samudra 2 captures the seasonal SST cycle, with warm piControl and cold 1pctCO₂ biases.



Seasonal SST variability transfers, while regional biases and anomaly errors limit zero-shot accuracy.

Reviewer concerns: target conditioning

Reviewer 3 · Major comment 3

“Dynamic Loss ablation. Main methodological contributions is the dynamic loss, but the other alternative evaluated are too simple and not reflective of current practice in SOTA models (i.e., computing loss in normalised space, using normalised time-delta targets (see Lam et al 2023)) which were designed to scale variables to the same range of value, and upweight slow changing variables.”

Reviewer 3 · Detailed comment 16

“Have you tried decomposing the prediction problem into predicting the trend, and the anomaly?”

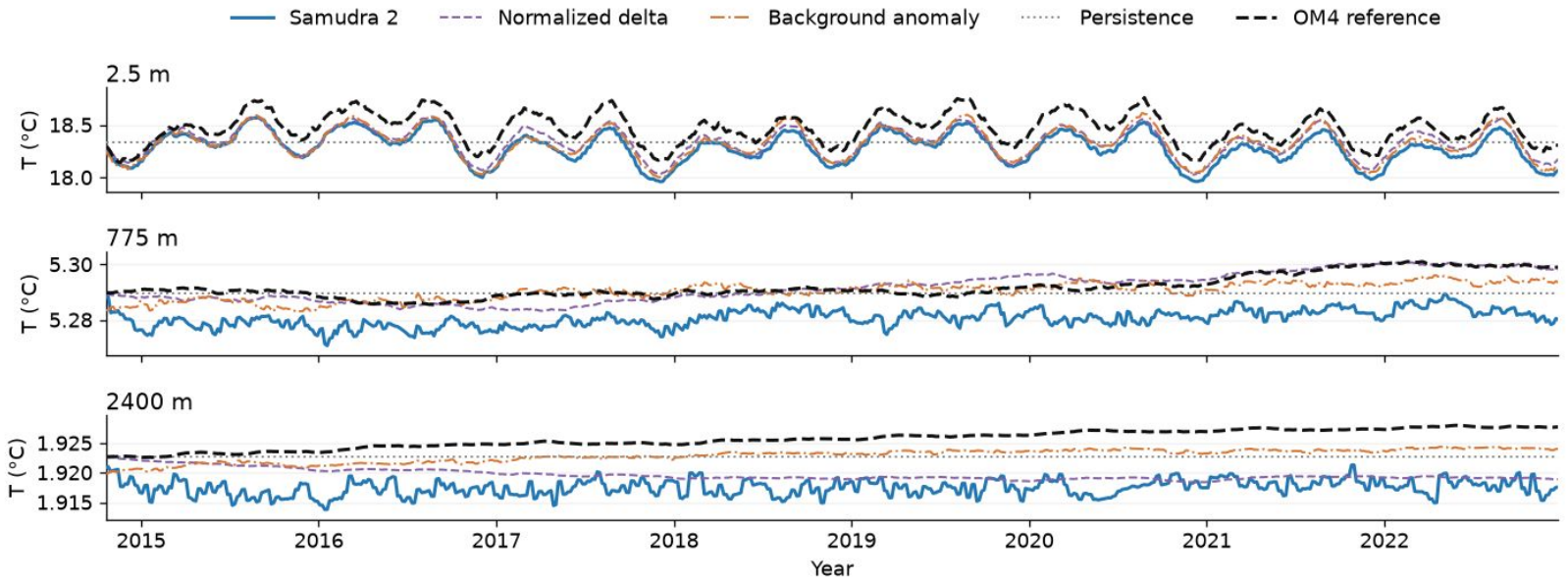
Models in the following comparisons

Model	Prediction target	How loss is weighted
Samudra 2	Full ocean state	Bounded inverse-error weights, then smoothed
Normalized-delta	Changes in ocean state	Fixed training-increment scales; no dynamic weights
Fixed-background anomaly	Anomalies + a fixed trend/seasonal background	Fixed training-residual scales; no dynamic weights
Frozen-final weights	Full ocean state	Final channel-loss weights, held fixed during training
Hu adaptive weights	Full ocean state	Smooth errors, then apply bounded inverse-error weights

Target tests: temperature

Both alternative targets reduce subsurface mean-temperature bias; surface cold bias remains.

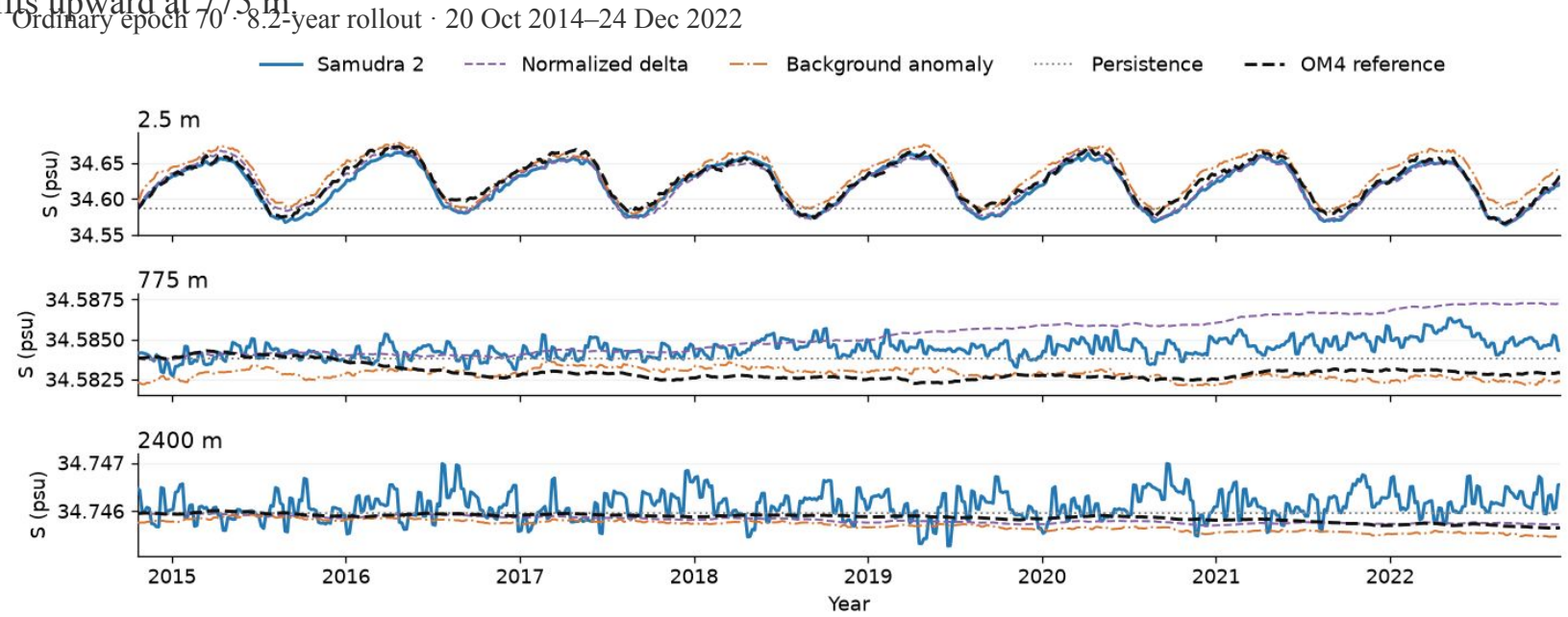
Anomaly: training-fitted trend/seasonality + learned residual. Delta: normalized increments.



Both alternative targets produce smoother subsurface mean trajectories in subsurface and deep ocean.

Target tests: salinity

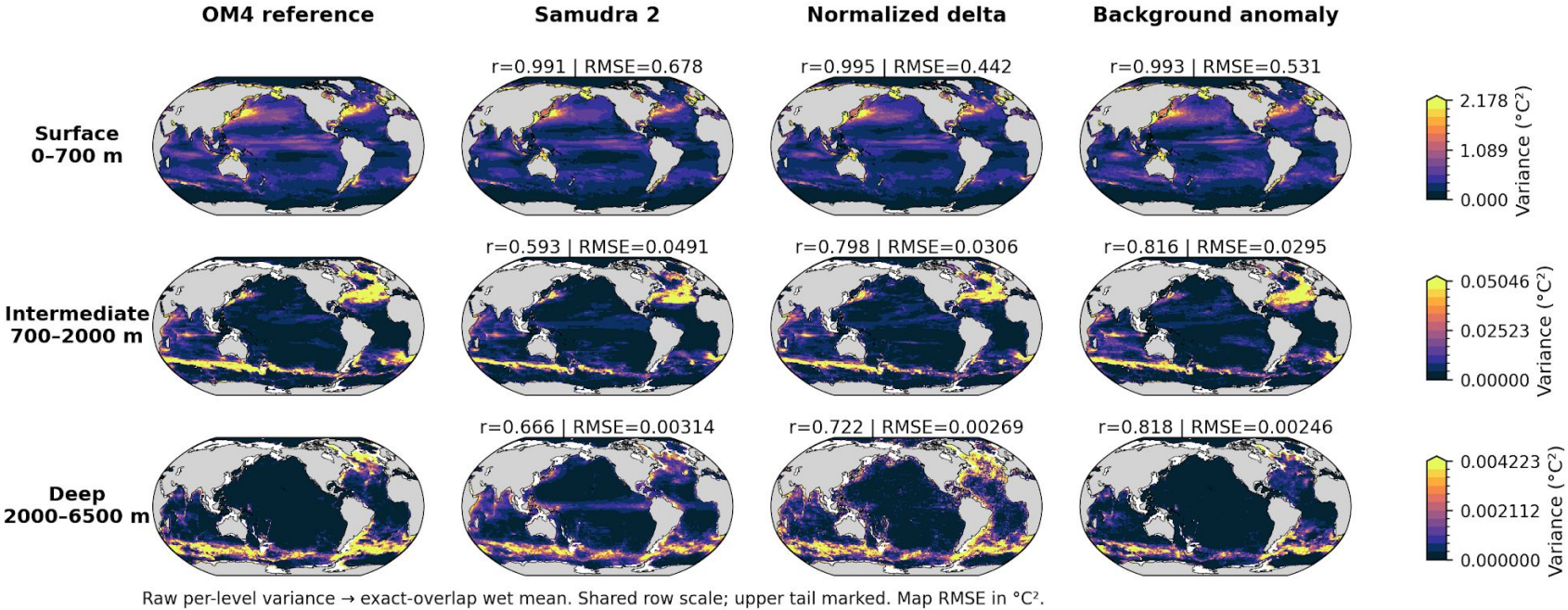
(check residual model) Anomaly prediction reduces subsurface mean-salinity bias; delta improves 2400 m but drifts upward at 775 m.



Background–anomaly prediction improves subsurface mean salinity and deep-band field accuracy.

Target tests: temperature variance by depth

Both alternative targets better reproduce temperature-variance patterns across all three depth bands.

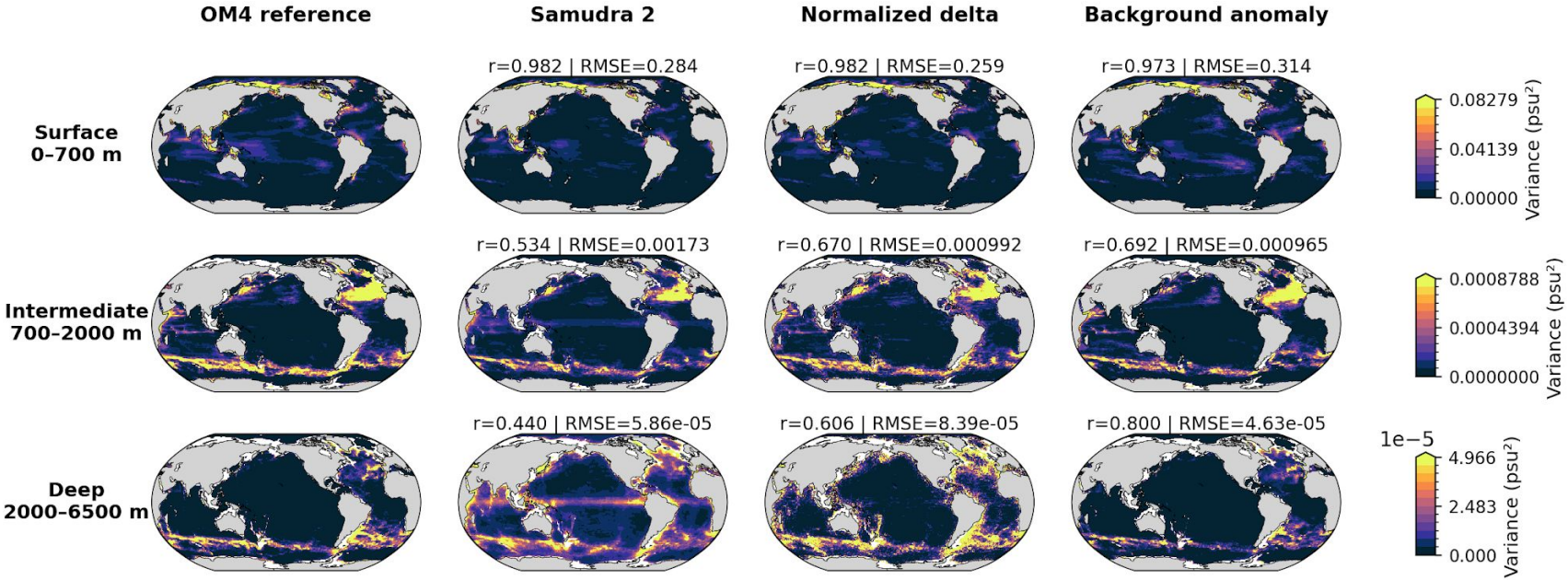


Raw per-level variance → exact-overlap wet mean. Shared row scale; upper tail marked. Map RMSE in °C².

Improved temperature-variance patterns coexist with overly weak deep variability in the anomaly model.

Target tests: salinity variance by depth

Both targets improve intermediate-ocean variance patterns; normalized delta exaggerates deep-salinity variability.



Raw per-level variance → exact-overlap wet mean. Shared row scale; upper tail marked. Map RMSE in psu².

Anomaly prediction better matches the deep-salinity variance pattern, but underestimates its overall magnitude.

Reviewer concerns: alternative weighting rules

Reviewer 2 · Major comment 1

“For example, Hu et al. (AAAI 2019) already used adaptive inverse-loss weighting, while GraphCast uses inverse temporal-difference variance to balance variables with different dynamical scales.”

“The authors should compare with these existing alternatives and clarify what is new about the proposed dynamic weighting.”

Reviewer 3 · Detailed comment 19

“1) use the loss weight values from the dynamic loss at the end of training(to understand if the benefit comes from the dynamic part, or simply the choice of values it converged to)”

Comparing alternatives and clarifying the method

Comparison

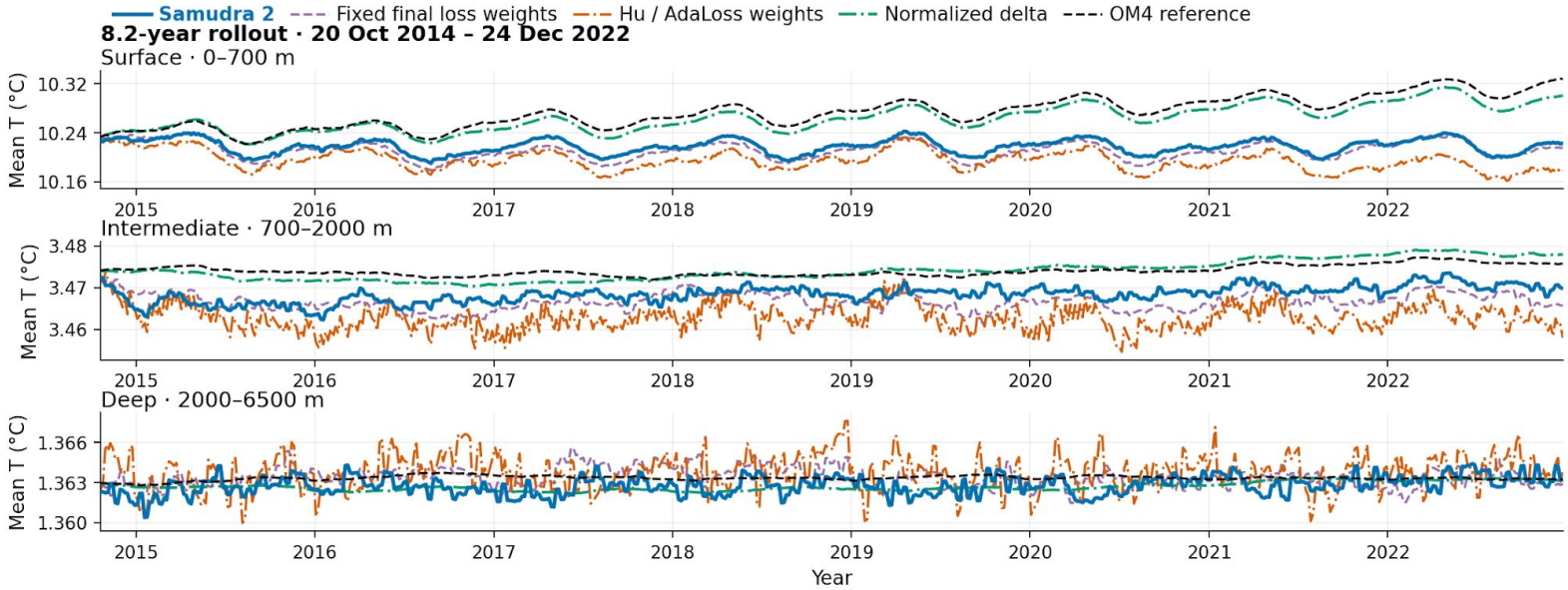
- Fixed final weights nearly match Samudra 2 temperature–salinity accuracy.
- Hu increases temperature–salinity field errors.
- Normalized delta has smoother mean trajectories but larger deep-ocean field RMSE.

Update rules

- GraphCast: FIXED scaling from historical increments.
- Hu: smooth errors, then apply normalized inverse-error weights with a floor.
- Samudra 2: form bounded inverse-error weights, then smooth the weights.

Loss controls: temperature

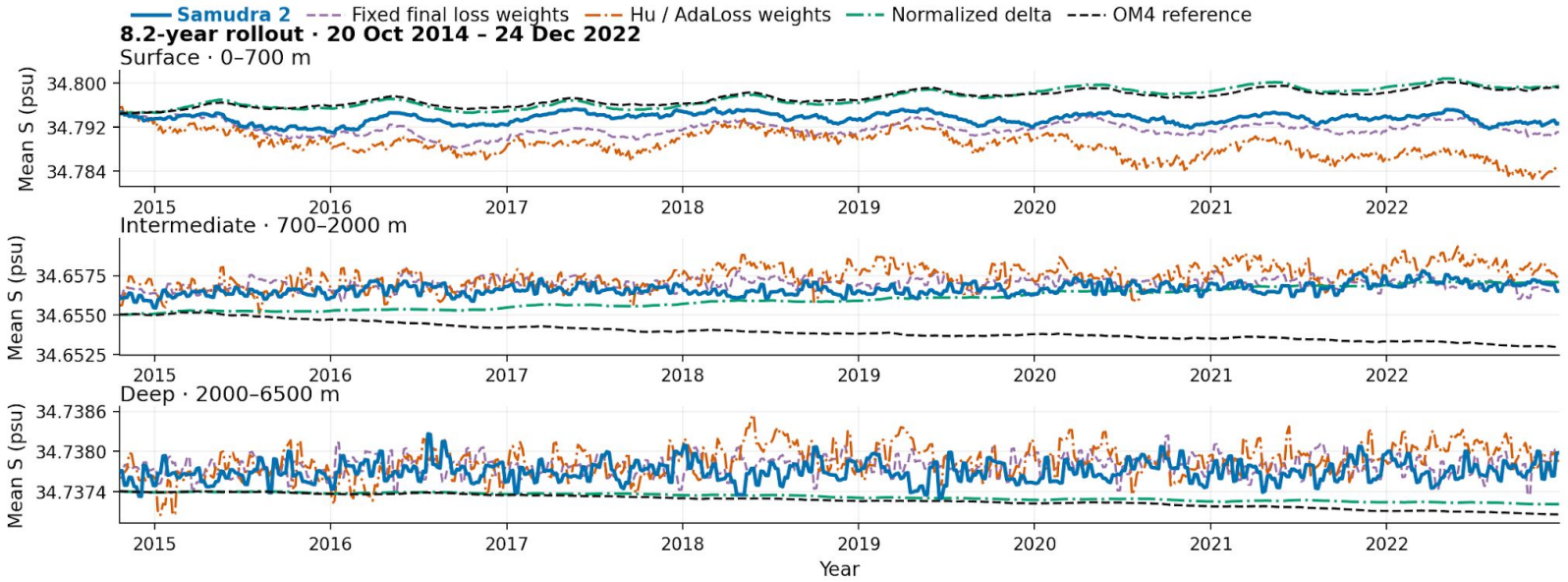
Fixed final weights produce similar temperature trajectories.



Fixed final weights retain most accuracy; dynamic weighting performs modestly better.

Loss controls: salinity

Normalized-delta prediction tracks mean salinity more closely, especially in the upper and deep ocean.



Reviewer concern: prescribed forcing

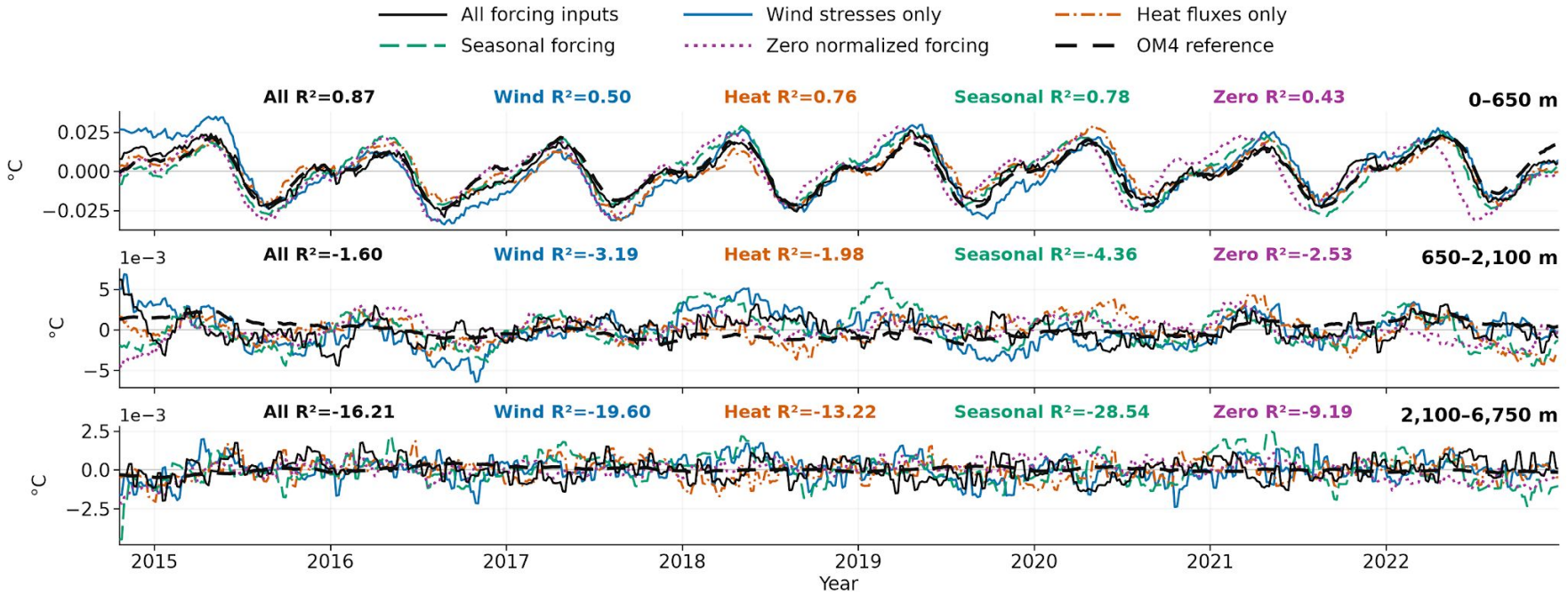
Reviewer 2 · Minor comment 1

“It is therefore unclear how much of the long-horizon skill comes from learning autonomous ocean dynamics and how much comes from continuously following the prescribed forcing trajectory. This is particularly relevant for upper-ocean and Niño 3.4 skill.”

“The authors should clarify the forcing provenance and, if possible, quantify its contribution through forcing ablations or sensitivity experiments.”

Forcing sensitivity: temperature

Full forcing best tracks upper-ocean mean temperature; deeper variability remains poorly reproduced.

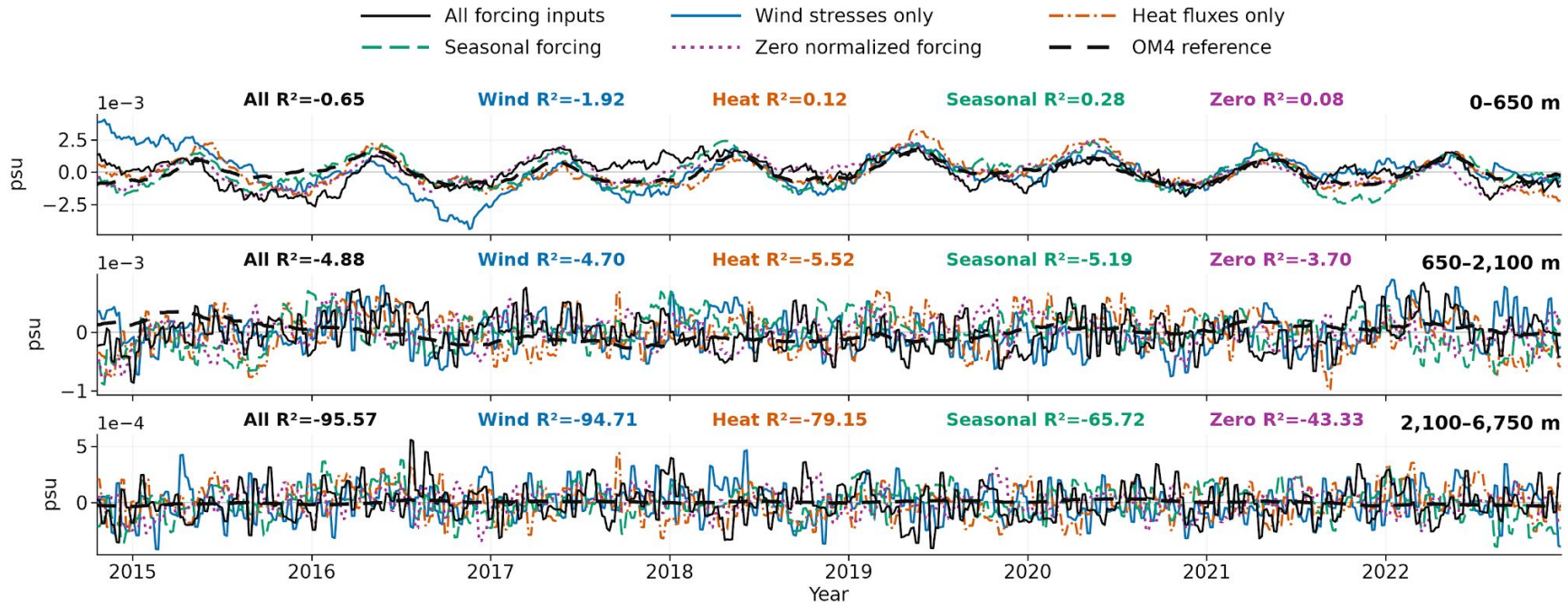


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Atmospheric forcing contributes to long-horizon upper-ocean temperature skill.

Forcing sensitivity: salinity

Seasonal forcing best tracks upper-ocean mean salinity; deeper variability remains poorly reproduced.
Random forcing instead zero



Prescribed forcing alone does not ensure realistic salinity variability.

Forcing sensitivity: Niño 3.4

Wind-only forcing slightly outperforms full forcing on these Niño metrics.

Forcing	Pearson r	Anomaly RMSE (°C)
Full forcing	0.967	0.222
Wind-only	0.971	0.213
Heat-only	0.863	0.484
Seasonal forcing	0.347	0.806
Zero normalized forcing	0.306	0.819

Time-varying wind forcing retains Niño skill.

Reviewer concerns: learning across channels

Reviewer 1 · Substantial comment 4

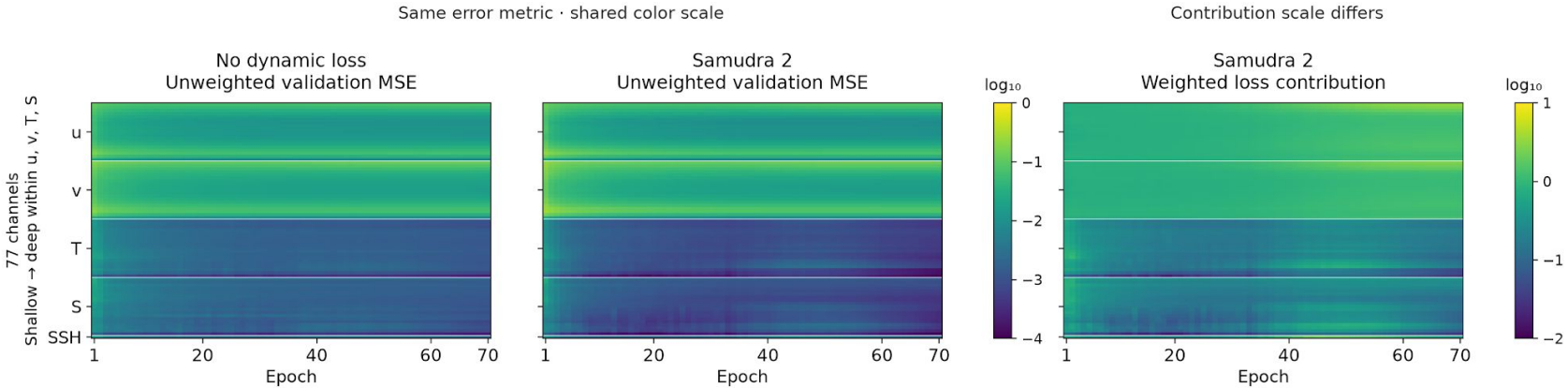
“It would be helpful if the authors could provide plots of the unweighted and weighted per-channel loss over the training steps on the training and validation set.”

Reviewer 1 · Substantial comment 5

“It would be good to show the training and validation curves to demonstrate whether there are signs of over-fitting to the training record in the sub-surface channels.”

Learning with and without dynamic loss

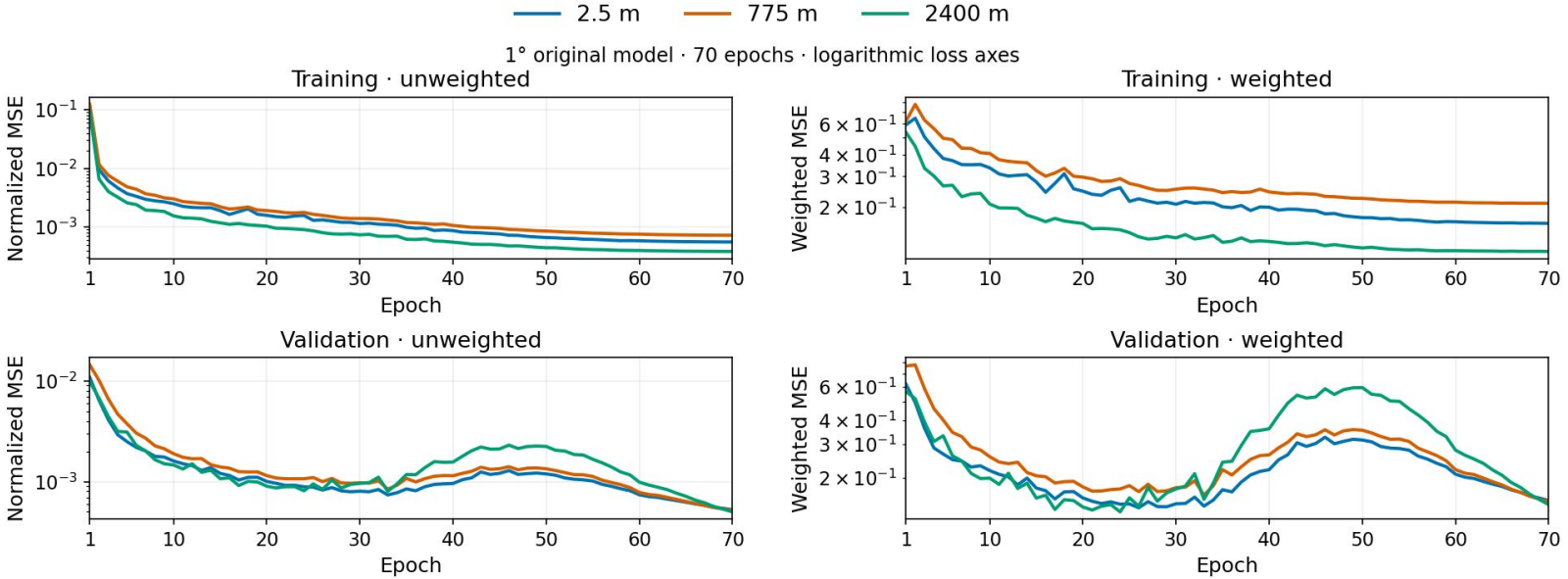
Held-out temperature and salinity errors are lower with Samudra 2 (epochs 61–70).



Dynamic weighting reduces the imbalance in channel contributions.
1° normalized validation MSE; historical comparison (upsampling differs).

Original-model learning curves: temperature

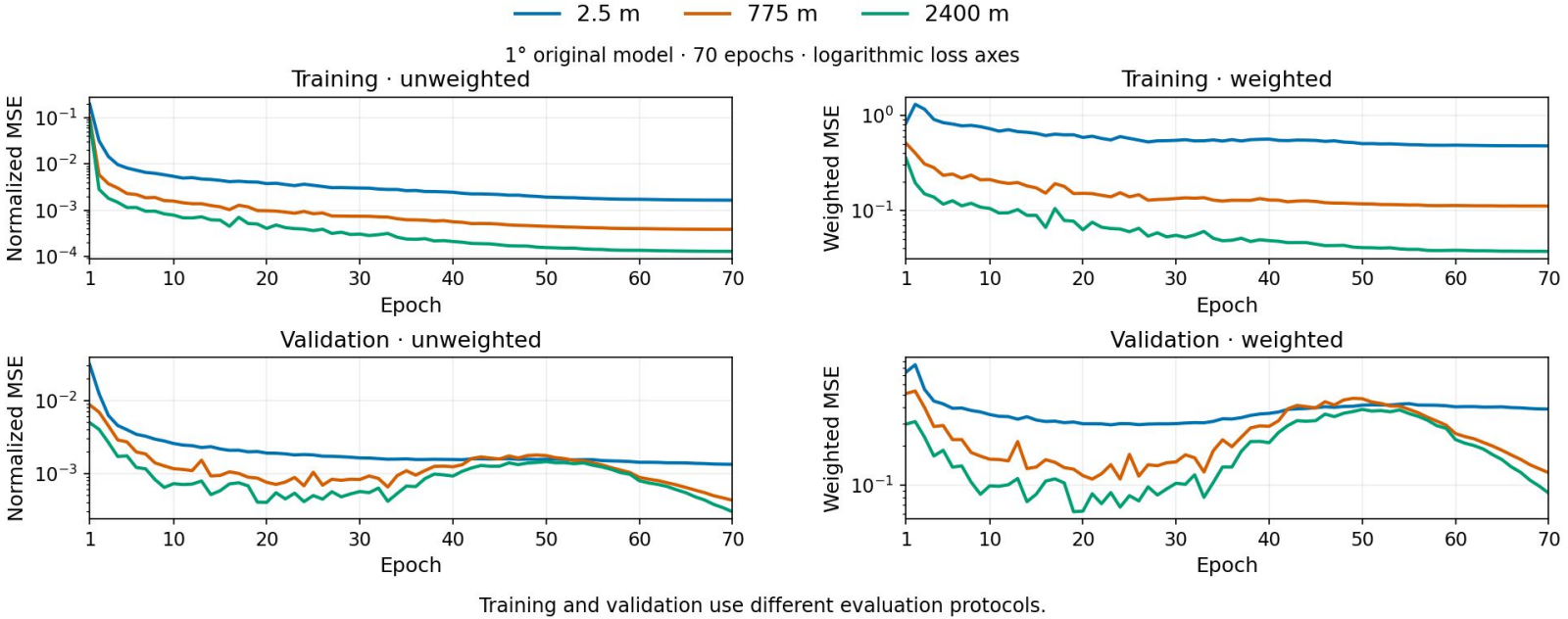
Deep-channel validation loss rises mid-training, then recovers by epoch 70.



Training and validation use different evaluation protocols.

Original-model learning curves: salinity

Subsurface salinity validation loss rises and then recovers; the surface behaves differently.

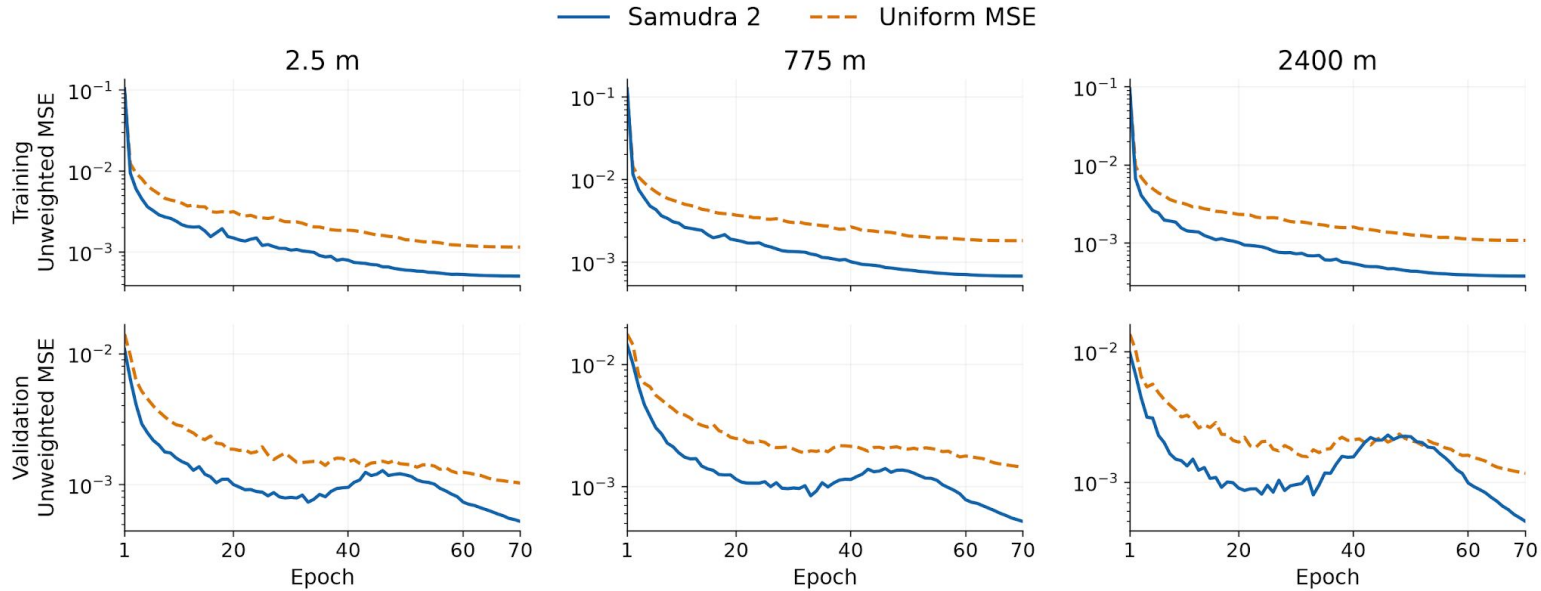


Training and validation use different evaluation protocols.

Overfitting checks should be channel-specific; a total-loss curve hides these differences.

Learning with and without dynamic loss: temperature

Validation MSE is 45–60% lower with Samudra 2 (epochs 61–70).
Historical comparison: upsampling also differs.

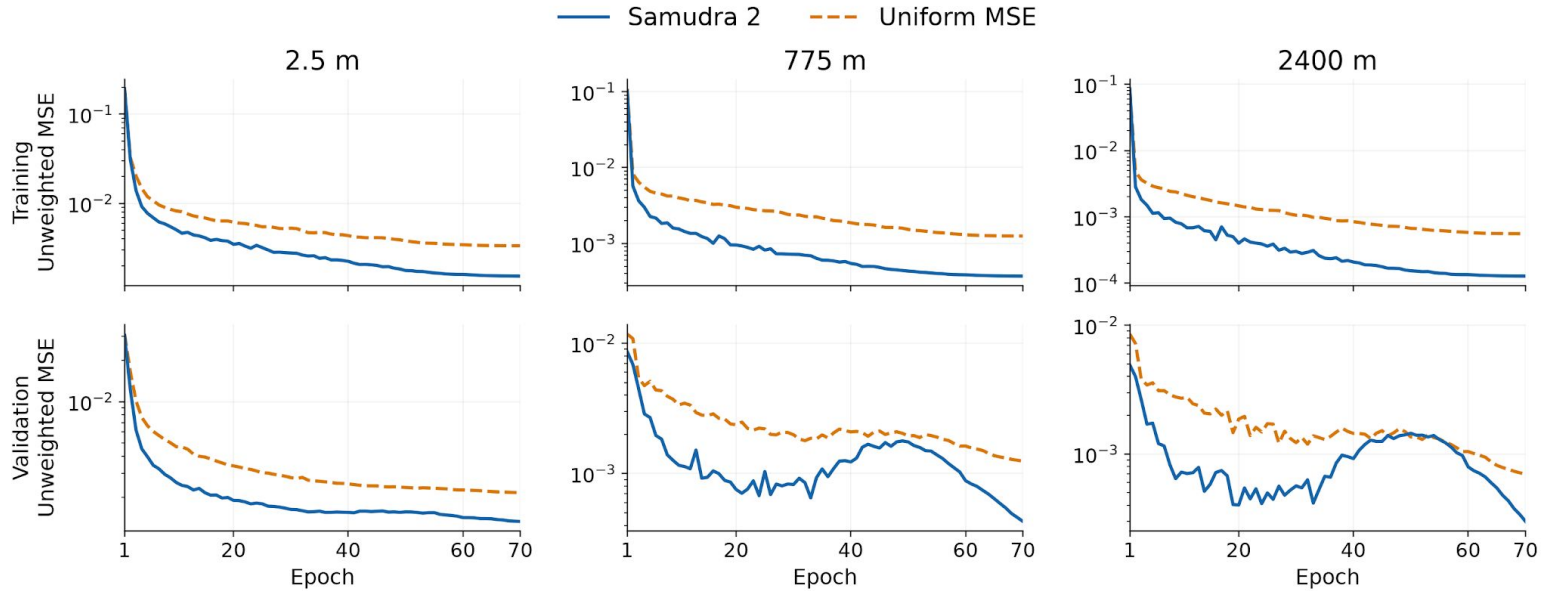


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Lower errors also appear in held-out validation at all three depths.

Learning with and without dynamic loss: salinity

Validation MSE is 38–55% lower with Samudra 2 (epochs 61–70).
Historical comparison: upsampling also differs.



Lower errors also appear in held-out validation at all three depths.

Reviewer concern: deep-ocean artifacts

Reviewer 2 · Major comment 6

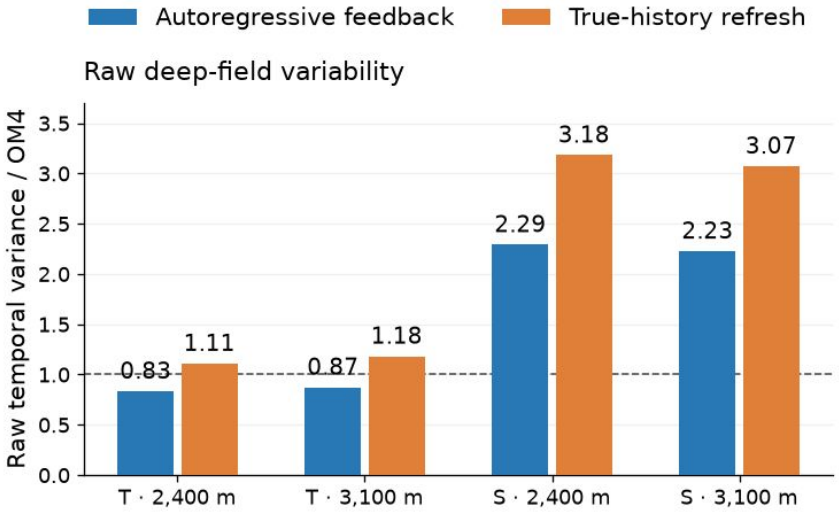
“Figures 5 and 6 show pronounced high-variance bands near the Equator in temperature and salinity, whose spatial patterns appear strikingly similar to the energetic KE structures in Figure 7.”

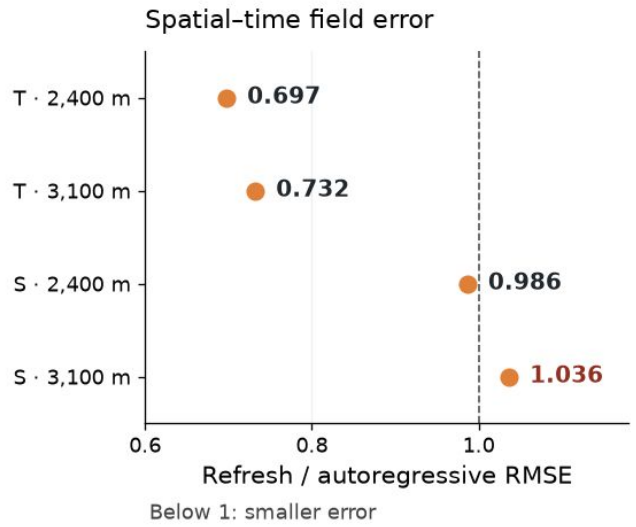
“The authors should discuss whether these features are physical or artificial, and investigate whether they arise from the shared architecture, loss design, training procedure, or preprocessing.”

Samudra 2: controlled feedback test

Refreshing the true history reduces most field errors, but excessive deep-salinity variability persists
Artifacts exist at one step.

Same checkpoint and forcing/numerical recipe · 598 outputs · 2014–2022





Long-rollout error accumulation alone cannot explain the deep-salinity artifact.

Deep-ocean controls: isolating input context

Two matched controls predict deep T/S with different ocean-state inputs.

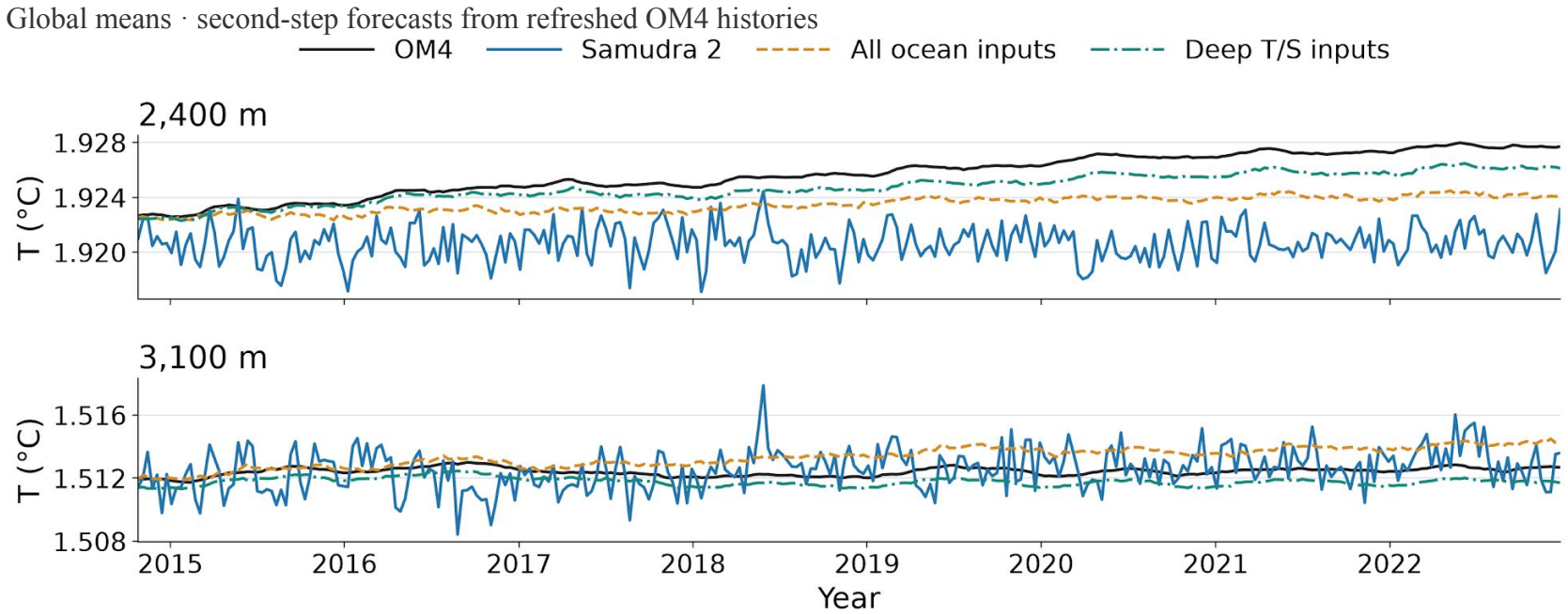
Model	Ocean-state inputs	Supervised outputs
Samudra 2	All ocean variables	All 77 state fields
All ocean inputs	All ocean variables	Deep T/S at five levels
Deep T/S inputs	Deep T/S at five levels	Same deep T/S fields

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Deep T/S inputs lower short-lead RMSE in all 10 fields versus both other models.

Deep-ocean controls: temperature time series

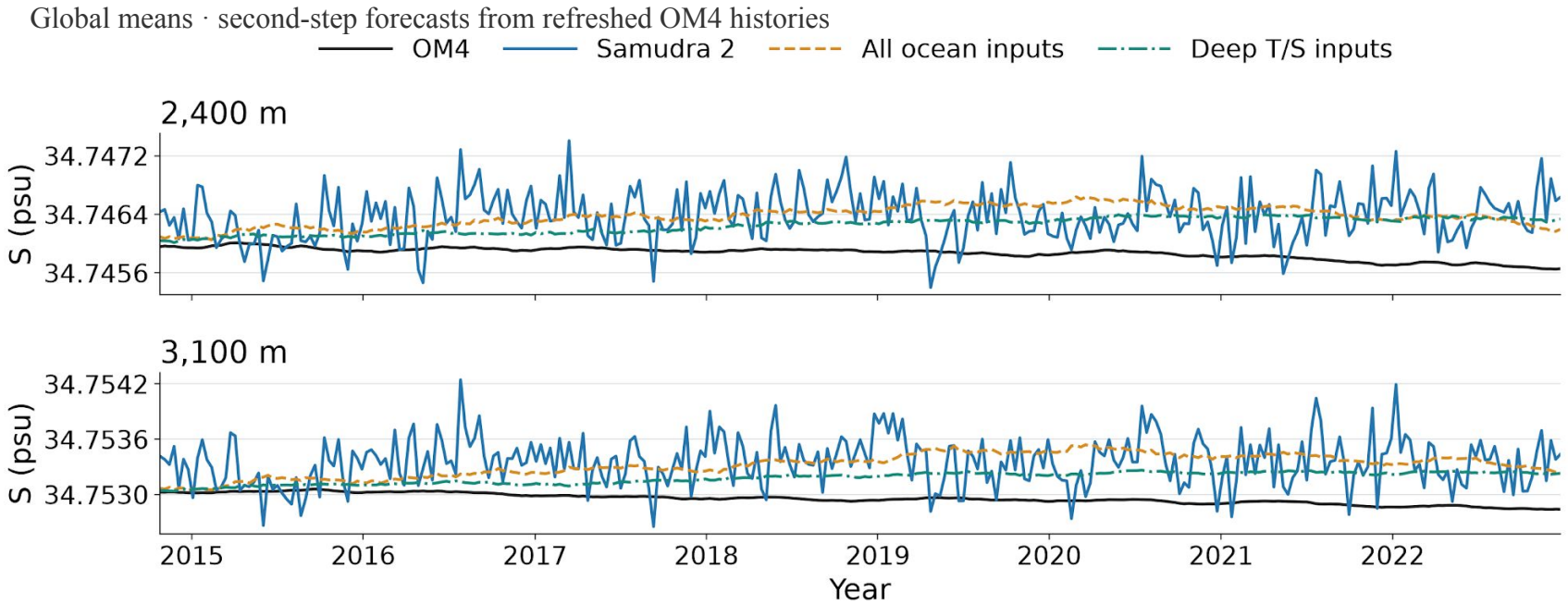
Deep T/S inputs give the lowest mean-temperature error at both depths.



Restricting inputs to deep T/S improves short-lead mean-temperature accuracy.

Deep-ocean controls: salinity time series

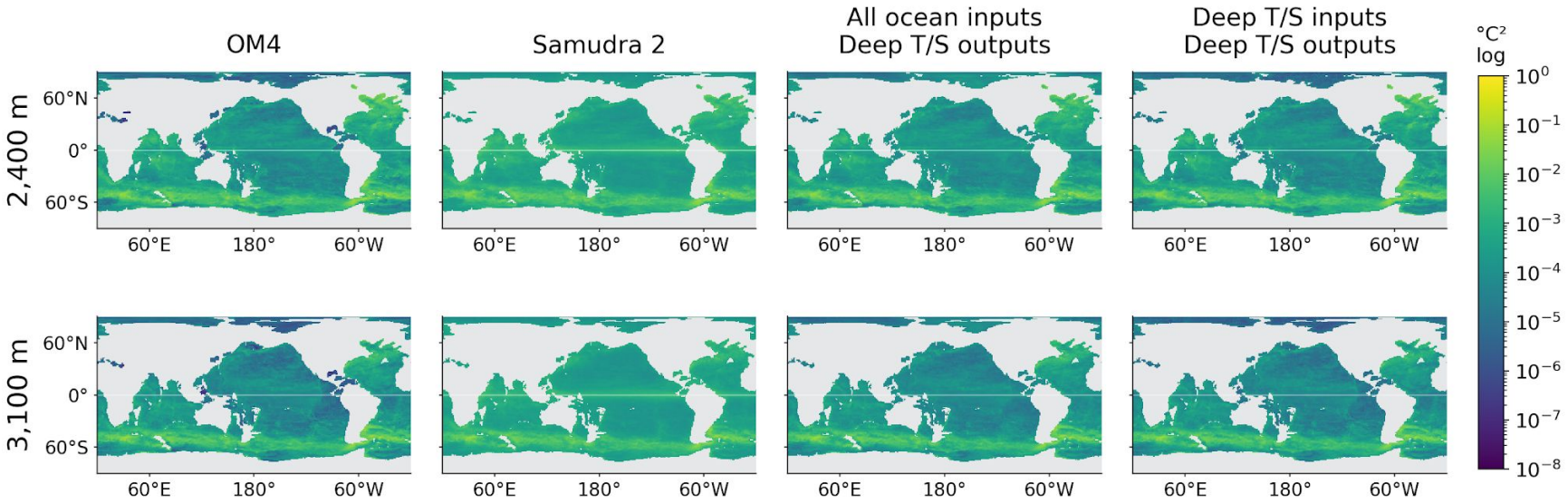
Deep T/S inputs reduce mean-salinity error at both depths; positive bias remains.



Input selection reduces, but does not eliminate, the mean-salinity bias.

Deep-ocean controls: temperature variance

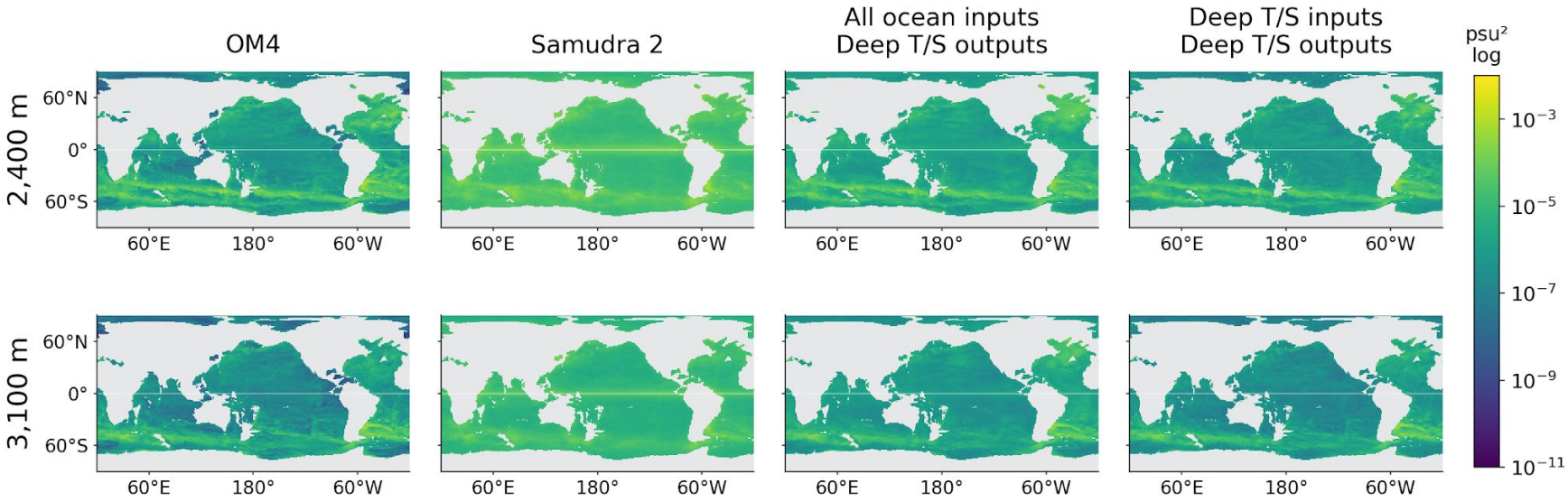
Restricting inputs to deep T/S further weakens the equatorial excess.



The matched input comparison implicates coupling to other ocean-state inputs.

Deep-ocean controls: salinity variance

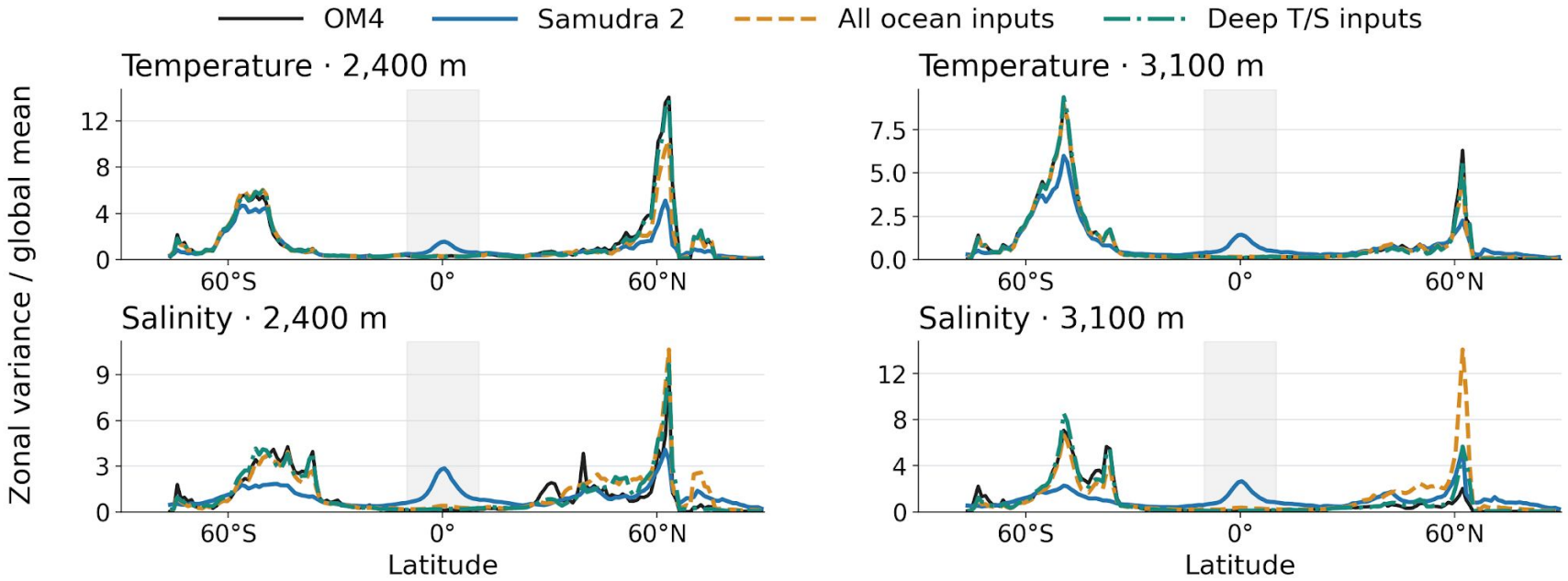
Both deep-output controls suppress the pronounced equatorial salinity band.



Deep T/S inputs retain only 59% / 77% of OM4 salinity variance at 2,400 / 3,100 m.

Deep-ocean controls: Where is deep-ocean variability concentrated?

The equatorial excess also weakens after removing each map's global amplitude.



Input restriction changes the spatial imprint, beyond uniform damping of variability.

Reviewer concerns: multiple starts and drift

Reviewer 1 · Substantial comment 3

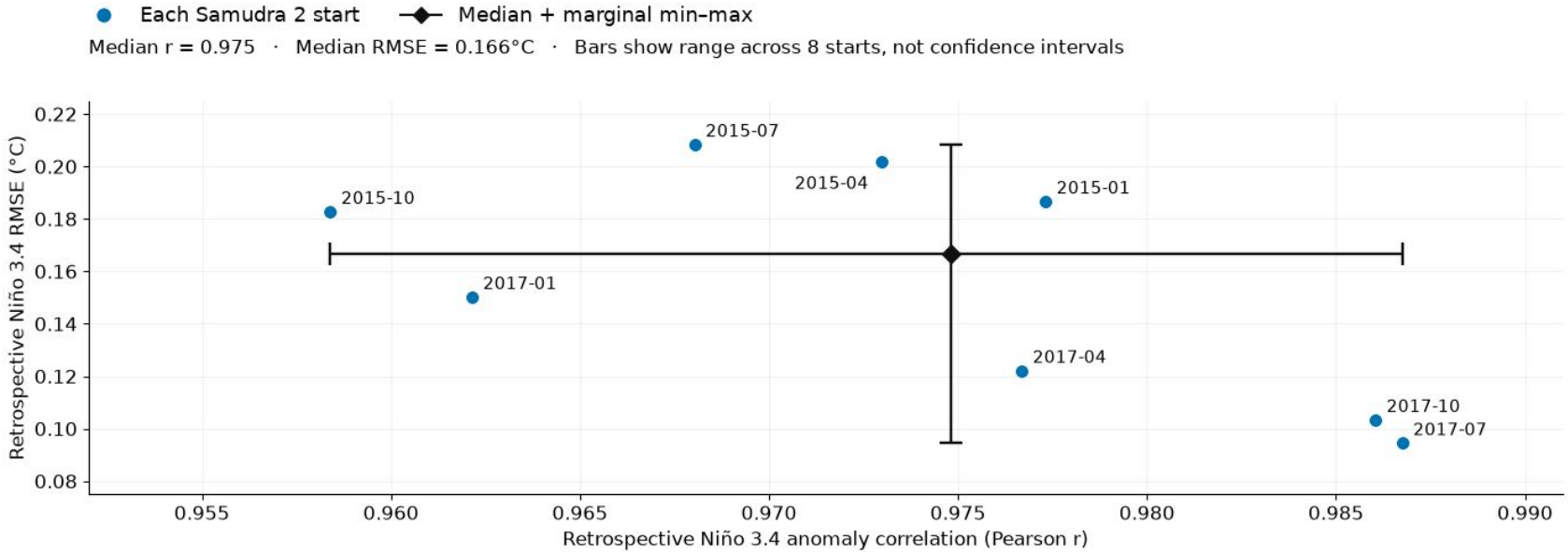
“It would improve the paper to include multiple rollouts from the model to show the variability in the output trajectories and possibly put some error bars on the numerical correlation values.”

Reviewer 1 · Substantial comment 6

“In an ocean emulator designed for climate models, it would be good to see some evidence that any drift in quantities such as global ocean heat content, global mean SSH look sensible compared to the GFDL model.”

8 historical starts (1°): Niño 3.4 variability

Niño 3.4 correlation stays high across all eight starts (r = 0.958–0.987).

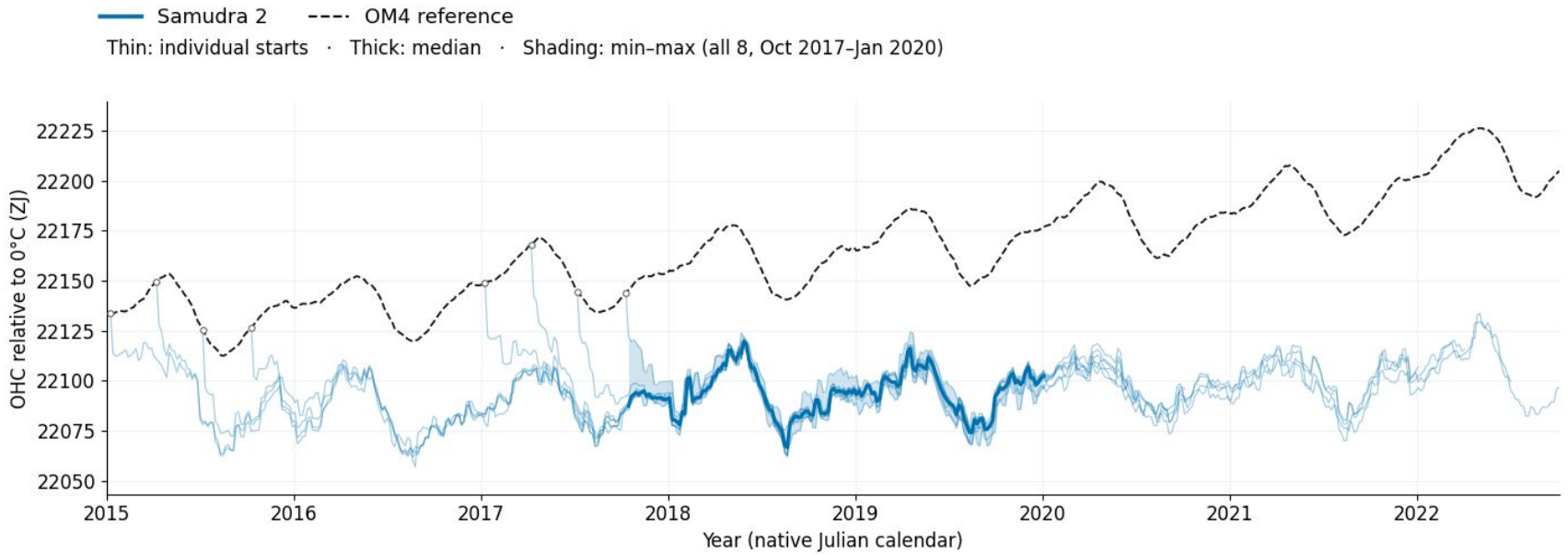


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Samudra 2 maintains consistent multi-year behavior across initializations.

8 historical starts (1°): ocean heat content

Eight initializations show a narrow OHC spread around a common trajectory.



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Across initializations, Samudra 2 shows similar trajectories but a consistent cold drift relative to OM4.